

Akhilesh Mahajan

University of Waterloo B.Sc. Candidate (2031)
Honors Physics (Co-Op Program) | 4.0 GPA

Driven to investigate fundamental questions in astrophysics
via observational data analysis and computational modeling.

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Skills

Programming	Python Packages	Software	Communication
MATLAB & Python C++ Java	NumPy Pandas & Seaborn Scikit-learn	Excel & Microsoft Suite AutoCAD Arduino & Circuit Simulation	Scientific Report Writing Research Poster Design Public Science Outreach

Projects

House Sales in King County, USA — Predictive Modelling | 2025

- Analyzed **21,000+ residential property records** using **Python** to clean, wrangle, and engineer features for predictive modelling.
- Built and compared **five regression models**, including Simple/Multiple Linear Regression, **Polynomial Pipeline**, and **Ridge Regression**, achieving **R² improvement** through regularization.
- Constructed a **scikit-learn Pipeline** automating preprocessing and prediction across **11 housing features**.

Automobile Sales Recession Impact Analysis — Data Visualization | 2026

- Designed **10 visualizations** using **Matplotlib, Seaborn, and Folium** to analyze automobile sales trends across **six U.S. recession periods**, covering GDP, unemployment, and consumer confidence.
- Built an **interactive choropleth map** and multi-chart dashboard identifying recession-driven shifts in **vehicle segment performance** and advertising strategy.

Double-Slit Interference Pattern Simulator | 2025

- Developed a numerical **wave-optics model** reproducing fringe spacing dependence on wavelength, slit separation, and distance.
- Incorporated **single-slit diffraction envelope** and **partial spatial coherence** effects.

Refracting Telescope - Optical Design & Fabrication | 2023

- Designed with first-principles geometric **optics calculations**. Built with **achromatic doublet lenses** and adjustable optical alignment.
- Verified performance against diffraction-limited resolution predictions. Observational validation via resolution of the moon.

Research Experience

Canadian Young Physicists' Tournament - National Competitor - Team Captain | 2025

- Conducted an original experimental investigation of neodymium magnet dynamics on rotating ferromagnetic rods
- **Designed experimental setup** to characterize system across controlled parameter sweeps of rotational velocity and friction regime.
- Developed competing theoretical frameworks, evaluated mechanisms using **quantitative experimental data fit comparisons**.
- Produced formal **research manuscript** meeting national competition standards. **Delivered technical defence** before physics faculty.

Diversity in Research Experiences Across Mathematics Institute - Competitive Research Immersion Programme | 2024

- Engaged in full **research workflow**: problem formulation → modelling → technical communication under faculty-mentorship
- Presented formal seminar on Lorentz transformations, special relativity, and **general relativity mathematical frameworks**.
- Collaborated with graduate student mentors and university researchers on modelling approaches and scientific interpretation.

Volunteer Experience

Peer Tutoring Programme for Students with Learning Differences — Spearhead & Lead | 2024-2025

- Provided individualized instruction in physics and mathematics **problem-solving**. Supported development of analytical reasoning.

Science Club, Don Mills Collegiate Institute — Founder & President | 2023–2025

- Established and scaled an organization to 20–30 active members. Organized school-wide **STEM events with 100+ participants**.
- Designed instructional programming in special relativity, wave optics, thermodynamics, and chaos theory.
- **Led volunteer team**, managing curriculum planning, demonstrations, and logistics.

Awards & Honours

Donna Strickland AMTD Award -- Scholarship Winner

Canadian Young Physicists Tournament – 6th Nationally

OAPT Physics Contest – School Champion, 2024

Academic Distinctions – Honour Role (95% Average)

Advanced Functions Award, Physics Award (Top Ranked)

Varsity Rugby Distinctions – Captain, MVP, 2022-2025

Educational Highlights

IBM Data Science Professional Certificate — Coursera | 2026

Hands-on training in statistical analysis, machine learning, and data-intensive computing applied to real-world datasets.

University of Waterloo — Minor in Computing & Astrophysics

Physics and computation-focused degree with hands-on experience in data analysis, scientific programming.